



everything is centralised within one hub and multiple tall stations can work in tandem in order to be able to process the data.

Transferring this data is another cumbersome task. Most drones come with a chip that is not allowed by secured systems, which makes the data transfer a big hassle. And let's not get started on the condition of 4G connectivity prevalent in the country.

Srivastava suggests opting for a cloud space where they can install and use software like ArcGIS. This helps in centralisation of the data without the need to transfer it from one system to another. Output can be monitored, and the 3D mine models can be viewed in real time. But then again, licensing for such software creates a hurdle, as everyone involved in approving a mine survey is always required to be in possession of it.

Srivastava points out, "There are few people who need to create, but there are many people who need to monitor; the GM of the area might like to see the mine. How does it look? He may like to see the land-use plans, but he may not have the software."

This is where the open source community can come to the rescue, as they create a middleware where the viewership can be done irrespective of whether one has that software or not.

Similarly, integration between the mining software and the point cloud data is also required. This would allow the point cloud data to be used on a software and allow the geologist to update their model from the real-time data, thus enabling a faster upgradation of models.

The other way is to make the model using a UTM bridge network where one can directly import the data into a common framework that is used by everyone. This makes monitoring easier for the regulatory bodies and even for the mining

bodies. It also becomes easier to integrate the mining software into the mine planning software, mine training software, and drones photogrammetry software.

Pai adds that emerging technologies like artificial intelligence and machine learning can also help accelerate how well-versed the hardware gets at solving major issues. He says, "While I don't think it completely eliminates the manual processes, because a surveyor on-site knows his mine better than anyone else, since drones capture visual 3-dimensional data, I think there is a tremendous scope of machine learning to automate a bunch of things."

### It all starts with skill development

While drone deployment in the mining industry has its share of hardware- and software-related challenges, a major obstacle that precedes them all is the lack of skillset related to drone technology amongst regulators and operators. It is especially prominent among surveyors, according to Srivastava.

Though it requires no qualification and limited knowledge of GIS software, drone operation is a skill that the industry is desperately looking for. Naturally, like it is with any nascent technology, it is much easier to get hold of young surveyors, but they still lack the eye that a veteran surveyor who has been in the field for a much longer period has. Training these surveyors is a win-win for the industry.

In the case of regulators, georeferencing becomes critically important. Learning the necessary software to analyse the drone data is necessary; an error in approval of land-use plans can have serious results.

Srivastava proposes the use of an open source SaaS model. These technologies cannot be used through local servers and laptops, and eve-

rything has to reside in the cloud. On the cloud, the licenses that are required will be different than the desktop licenses, and integration of mining software and the photogrammetry software requires more work to be done.

### What the future looks like

Recent developments in the drone industry in India have brought about a huge opportunity for simplifying the mining process. The vision is in place, though the execution is still slow.

Sahu says, "I would like to see drones being used on a daily basis from surveillance to utilisation and monitoring of logistics, where there is enough room. The technology needs to be developed and taken to a level where continuous work is done and another tech can come in to make it robust."

He adds, "I would like to see where discoveries are being made using drones, because I strongly believe that is one area where it can play a vital role, and new mineral discoveries can be made using drones sensibly and wisely." Exploration through drones has definitely picked up the pace, but it is yet to reach the stage where the benefits can outweigh the benefits of other traditional exploration means.

Drone solutions providers need to address the issues quickly. While the golden period of drones use has just begun in the country, this is the perfect time for their use in a sector that is expected to grow at an unprecedented pace. With good research and collaboration, the scope of drone solution providers being able to provide scalable solutions that the mining sector desperately needs increases manifold, thus increasing the possibility of progress in the future. **EFY**

*This article, transcribed and curated by EFY's Siddha Dhar, is based on the panel discussion on 'Drones In Mining' that was hosted recently by ASSOCHAM.*